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# EE1060: Differential Equations and Transform Techniques

## Lecture 37: Homogeneous Systems of Differential Equations

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### Topics :

1. Solution of Homogeneous System of Differential Equations
  2. Long Term Behavior - Phase Portrait
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## Homogeneous System of Differential Equations - Construction of Fundamental Set

given a set of homogeneous first-order ODE

$$\dot{\vec{x}} = A\vec{x} \quad \text{where } \vec{x} \text{ is of length } n. \quad \text{--- (1)}$$

Construction of fundamental set:- for an ODE

$$\left. \begin{aligned} a_n \frac{d^n x}{dt^n} + \dots + a_0 x &= 0 \quad \text{--- (2)} \\ \text{Solution : } x &= e^{\lambda t} \quad \text{--- (3)} \end{aligned} \right\} \text{from previous discussions}$$

when (2) is reformulated as (1):  $\vec{x} = \begin{bmatrix} x \\ \dot{x} \\ \vdots \\ x^{(n-1)} \end{bmatrix}$

based on (3), we can pick a trial }  $\vec{x} = \underbrace{\begin{bmatrix} 1 \\ \lambda \\ \vdots \\ \lambda^{n-1} \end{bmatrix}}_{\vec{v}} e^{\lambda t}$   
 solution for (1)

So the trial solution for  $\dot{\vec{x}} = A\vec{x}$  will be  $\vec{x} = \vec{v} e^{\lambda t} \quad \text{--- (4)}$

if we can construct  $n$  solutions of the form (4)  $\rightarrow$  Construction of the fundamental set.

## Construction of Fundamental Set - Simple Case

for the trial function  $\vec{x} = \vec{v} e^{\lambda t} \rightarrow (4)$  to be a solution, it must satisfy  $\dot{\vec{x}} = A \vec{x}$

plug in (4) in (1)  $\Rightarrow \dot{\vec{x}} = A \vec{x}$

$$\lambda \vec{v} e^{\lambda t} = A \vec{v} e^{\lambda t}$$

for all choice of  $\lambda, \vec{v}$  such that  $A \vec{v} = \lambda \vec{v}$ , Eq (4) will be a sol to (1)  
 $\rightarrow$  eigen value problem.

(5)  $\Rightarrow \vec{v}$  is in the null space of  $(A - \lambda I)$

$\Rightarrow \det(A - \lambda I) = 0 \Rightarrow$  characteristic polynomial  
 $\downarrow$  Sol  
 eigen values ' $\lambda$ '.

eigen vector corresponding to eigen value  $\lambda \Rightarrow (A - \lambda I) \vec{v} = 0$ .

fundamental set of solutions for  $\dot{\vec{x}} = A \vec{x}$  will have elements of the form  $\vec{v} e^{\lambda t}$  where  $\lambda$  is the eigenvalue &  $\vec{v}$  is the corresponding eigen vector.

Main Question :- if we can construct 'n' such vectors of the form (4), then the construction of fundamental set is complete. But can we construct them?

## Construction of Fundamental Set - Simple Case

Need to consider 3 scenarios

Case I:- 'A' has n distinct real eigen values.

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Corresponding eigen vectors are linearly independent.

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fundamental set of sol are  $\{\vec{v}_i e^{\lambda_i t}\}_{i=1}^n$

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General sol:  $\vec{x} = \sum_{i=1}^n \alpha_i \vec{v}_i e^{\lambda_i t}$

$$\text{or } \vec{x} = \underbrace{[\vec{x}_1, \dots, \vec{x}_n]}_{\text{matrix } X} \underbrace{\begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix}}_{\text{vec } \vec{\alpha}}$$

$$\text{gen sol :- } \vec{x} = X \vec{\alpha}$$

for an Ivp  $\dot{\vec{x}} = A\vec{x}$  and  $\vec{x}(t_0) = \vec{x}_0$ ; find gen sol & compute  $\vec{\alpha}$  based on Init Cond.

To be discussed later.

Case II:- A has repeated eigen values.

Case III:- eigen values of A come in complex conjugate pairs

## Example

Compute the solution of  $\dot{\vec{x}} = A\vec{x}$  where  $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$  and init Cond  $\vec{x}(t=0) = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$

Eigenvalue & eigenvector pairs of  $A$  are  $(\lambda_1 = 2, \vec{v}_1 = \begin{bmatrix} -1 \\ 2 \end{bmatrix})$

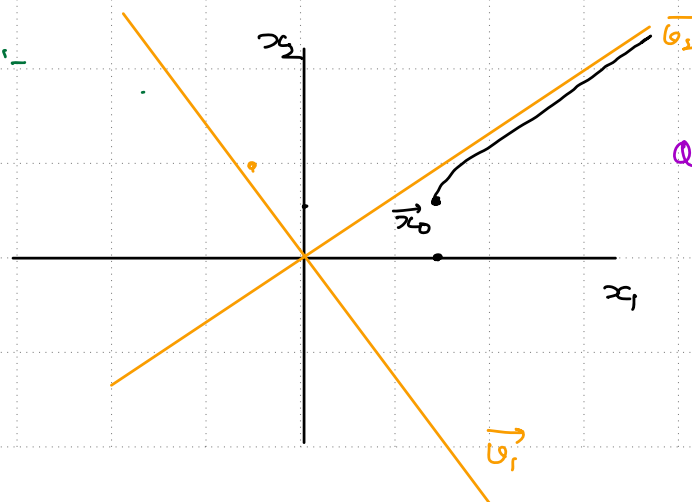
$(\lambda_2 = 5, \vec{v}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix})$

general solution  $\vec{x} = \alpha_1 \vec{v}_1 e^{2t} + \alpha_2 \vec{v}_2 e^{5t}$

for the given initial condition  $\rightarrow \alpha_1 = -4/3$  and  $\alpha_2 = 5/3$

solution to the given IVP is  $\vec{x} = (-4/3 e^{2t}) \vec{v}_1 + (5/3 e^{5t}) \vec{v}_2$

Trajectories or Phase-portrait:-



Question: what happens  
as  $t \rightarrow \infty$ ?

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To be discussed in  
Lec 38.